D596 Task 1: The Data Analytics Life Cycle

For Task 1 of D596: The Data Analytics Journey, I will demonstrate my knowledge of the Data Analytics Life Cycle (DALC) across three main points below (A, B, C). Each has subpoints (A1, A2, B1, B2, C1, C2, C3) to provide more information and further demonstrate my knowledge of the DALC based on my own experience. I will highlight areas that require more expertise and form an action plan to gain more experience. Finally, I will select and explain how a data analyst might use a selected tool or technique in an organizational setting and the pros and cons of that tool or technique, along with ethical concerns.

The Data Analytics Life Cycle

A. Describe each of the seven phases of the data analytics life cycle, adding a reflection of your own expertise to each phase described.

The following are the seven phases of the data analytics life cycle (DALC).

1. Business Understanding
    a. The Business Understanding or planning phase is the first phase data analysts often begin with when starting a new project. Analysts will meet with stakeholders during the business understanding phase to better understand the business goals and objectives. During these meetings, analysts will seek to answer and ask relevant questions related to the project to identify access requirements, necessary resources, constraints, and roadblocks and determine a timeline. A lack of understanding during this phase can result in going over budget, not meeting deliverable timelines, or the complete failure of the project. **Reflection**: With 10 years of experience as a business analyst, this phase is also where I usually start my projects. Understanding the business goals and stakeholder expectations before beginning any project's heavy lifting is essential; otherwise, one risks not meeting stakeholder expectations and creating more work for the team. As the analyst, stakeholders often consider you the subject matter expert (SME) and expect you and your team to have their best interests in mind when planning a project. Part of this is identifying risk, asking stakeholders the right questions to clear any fog around their request, and setting realistic expectations for all parties involved.
2. Data Acquisition

   a. During the data acquisition phase of the DACL, data analysts work to acquire the necessary data to accomplish their analysis goals. This can come from many sources, such as SQL or No-SQL databases with tools like Microsoft SQL Server, PostgreSQL, or MongoDB. All necessary data would be available through a database in a perfect world. However, that is not always the case, and sometimes analysts may need to be creative in sourcing their data with tools such as Python and performing web scraping or even just the old-school methods of surveys through tools like Survey Monkey or Google Forms. The use of APIs and building data pipelines to perform full extract and transform load (ETL) processes can also occur during this phase, as some data sources may require this added work to access.

      Reflection: In my current role as a business analyst, data acquisition is typical, often involving sourcing test data from our database utilizing SQL queries to develop test cases, confirm test results, or verify environment configuration. This data can range from customer, account, transaction, or alert information the system passes through data pipelines to tools like Kafka.

3. Data Cleaning
   a. One of the most critical phases of the DALC, as errors in this phase can result in extra work down the road or invalid results in your reporting and visualizations. In this stage, analysts will use various tools such as SQL, Python, R, or even Excel to transform and cleanse the data in preparation for the next steps. The analyst's goal is to ensure that the data going is relevant and clear of any errors, such as invalid values, duplicate records, inconsistencies in formatting, too many null values, and outliers.

Reflection: Data cleaning is not as standard in my current role as data validation. While triaging some test environment issues, I modified account database records through SQL scripts. Either through inserting entirely new records, modifying select values for a record, or even dropping entire rows of records when we discovered our ETL process was filling our database with duplicate values in error.

4. Data Exploration

    a. Exploratory data analysis (EDA) allows data analysts to understand better the data they are working with and identify the relationships between the different datasets they may use. The data exploration phase is when data analysts want to utilize data visualization tools and numerical summaries to discover patterns and identify measures of center, variability, and distributions within the dataset.

    Reflection: In my current role, I participate in the annual tax process, which involves testing the new tax code release in multiple lower test environments. As part of this process, we run our yearly tax jobs, which load our tax database tables through a separate ETL process. As part of the test process, I must explore the tax database using SQL to confirm that the tax batch process loaded the customer data correctly and that the fields contain the expected data. I will review row counts and validate that all expected columns are present. While confirming the data is valid, there are no null values in specific fields like the customer’s name, SSN/TIN, address, or phone, and those values are in the correct format. If I notice an anomaly in the data, I then move to analyze the data for patterns that may be informative for future triage sessions.

5. Predictive Modeling

a. In this phase, the analyst will create data models, correlation-based models, regression models, and time series they will use to make predictions using the dataset. While designing these models, analysts will want to select the appropriate model for their dataset. For example, a time series model requires enough time data to be precise. The analyst will also need to cross-validate their predictive models for accuracy. To help train these models, data analysts will utilize Python or R to automate the training process. Allowing for an overall more accurate data model.

   Reflection: The most similar task to predictive modeling I have performed in my current role is analyzing trends in Jira tickets logged and the time it has taken to resolve those Jira tickets. For example, I track my performance in one of my dashboards to monitor my average resolution time for tickets. This dashboard provides me with actionable Key Performance Indicators (KPIs) to base my performance. Using similar dashboards, I track Jira issues related to our tax system, which allows me to gauge how many issues related to our tax system are being raised and how long it takes our engineers to provide a solution. My team utilizes this dashboard to track the problems throughout the year, identify critical issues, and predict when our engineers will provide resolutions.

6. Data Mining/Machine Learning

   a. In this phase, data analysts will create smaller training datasets based on the original dataset to build models, identify patterns and clusters if they exist, and classify the data. Then, these smaller training data subsets help train and test against data models. Tools like Python or R play a key role in allowing the data analyst to automate the training of their data models.

Reflection: Data mining is not a process in my daily workflow and would be difficult to incorporate. I could gain experience in this step through portfolio projects like building a small machine-learning model that offers musical artist recommendations based on a user's past ratings of other musical artists. This would require identifying patterns and predicting what the user will find enjoyable.

7. Reporting and Visualization

    a. The final step in the DALC is where the data analyst finally utilizes the data they have collected and analyzed in the previous steps to tell the stakeholders/audience a story. They will summarize their analysis and provide helpful insight through graphs demonstrating trends and forecasting stakeholders' use for business decisions. Data analysts may also create dashboards at this point to allow stakeholders a more direct view of the data. Allowing the data to be updated moving forward through set refresh schedules of the dashboard.

    Reflection: Currently, I present data through Jira visualizations and occasionally Excel spreadsheets to small groups within my direct line of business. For example, in our bi-weekly tax defect meeting, I present a Jira dashboard I built for tax-related Jira issues and report our current metrics. The dashboard selects tickets based on specific selection criteria. It displays multiple metrics such as counts by severity, ticket summary with status, time in status, time since the last update, and the latest comment, to name a few.

A1. Propose a way, with at least one example of each, that you might gain expertise in each of the seven phases.

Over the years, in my current role as a business analyst, I have gained experience in some phases of the DALC process by working on many internal and external projects with a broad scope and business impact. As I focus on moving into a data analyst role, I will gain firsthand experience in the job role. Before acquiring this position, I can gain more experience in business understanding by communicating with my manager my desire to be more involved in stakeholder meetings. One option is shadowing my manager in upper management and client-facing meetings to understand the business goals at a higher level. For data acquisition, I will continue to seek opportunities within my current role to utilize SQL to collect data when needed and find more Udemy and other courses on web scraping with Python. Data cleaning is a phase I will need to use external courses and this master's course to gain more experience in the process since my current role does not offer a chance to explore this phase in depth. Data exploration is another phase I perform in my current role. I can gain some experience by diving into our databases to seek answers. However, performing EDA on different types of data can be beneficial. I will seek out available courses that offer datasets and utilize free datasets online to perform portfolio projects to bolster my resume for future employment. I have created predictive models utilizing online courses, but I need more experience in this phase. Besides what we learn in the MSDA program, I plan to use more online courses and develop my training and predictive models for my portfolio. Data mining is the phase I utilize the least in my current role and would have the most challenging time implementing. However, I have Udemy courses that specialize in data mining, and I plan to complete them and use the knowledge gained in the master's program to perform my data mining as part of my projects. I have experience with creating reports and visualizations with tools like Tableau. Gaining more experience in my current role with tools like Tableau may be difficult. However, I can continue creating dashboards for our different project efforts and, in my own time, develop visualizations through Tableau for personal projects and online coursework.

A2. Explain how the goal and mission of the organization help the analyst to identify the business requirement.

The current goal of my organization, Release Management, is to convert all current clients to our latest software version and unify our codebase by the end of 2025. We aim to streamline support across all clients, increase overall client satisfaction ratings, and reduce the workload of our developers. This mission sets a clear business requirement and direction for analysts to prioritize tracking client upgrade progress through tools like Jira and to create useful dashboards and visualizations for stakeholders utilizing Tableau or Power BI. With our goal of unifying our codebase into a single code line by the end of 2025, the data analysts can use data on current client versions – base code, release, or hotfix version. – or current resolution times on logged issues to create actionable suggestions based on the organizational need for release management that aligns with the end goal.

Data Analytics Tool or Technique

B. Apply your knowledge of the data analytics life cycle by selecting one data analytics tool or technique and describing how the tool or method might be used in one phase of the data analytics life cycle in an organization about which you have some knowledge.

Across many phases of the DALC, data analysts can find a use for SQL. However, we will focus on using SQL in the data acquisition phase. In the data acquisition phase, SQL can collect and retrieve custom datasets from databases by executing custom queries and sub-queries. For example, an analyst at Walmart may run a SQL query against a relational database to obtain sales data they would use to create sales forecasting models. Leveraging SQL in the data acquisition phase allows the analyst to obtain relevant data at scale to provide actionable data for stakeholders.

B1. Include three risks of using the selected tool or technique for data analytics.

Three risks of using SQL as a data analytics tool:

1. Data Security
    a. A business should practice proper sanitization standards, or the business's SQL databases may be vulnerable to SQL injection attacks. For example, let's say an analyst goes to an online form to ask for help with an SQL script they cannot get to execute. A random forum user provides the analyst with a revised script, and the analyst takes the script and runs it against the business's database. The script could contain malicious lines to modify or delete sensitive data and compromise the database or, worse, the organization's security.
2. Data Quality

a. Databases can contain data of questionable accuracy or quality due to the nature of which the data was created and stored. Using SQL to query this data does not magically clean the data and remove issues such as incomplete, inconsistent, or erroneous data. Even if the database is perfect, the SQL query is still subject to human error. Targeting or joining the wrong tables or having missed WHERE clauses can produce inaccurate results in a SQL query. An analyst might not notice these errors, resulting in the analyst providing incomplete or misleading data to stakeholders.

3. Lack of flexibility
    a. SQL can be a powerful tool; however, it changes when analysts need to interact with unstructured or semi-structured data (e.g., text files, JSON). Data such as this will limit the usefulness of SQL to data analysts if the data is only available in these formats.

B2. Describe an organizational or technical problem using the selected tool or technique.

My current organization has revamped our database schema multiple times to support newer environment requirements for moving to a multi-tenant(client) test environment. This change has broken many of our existing ad hoc SQL scripts, queries, and automated batch scripts. This defect created unfortunate delays in our deployment process, as the required batch jobs utilizing SQL scripts took hours to finish. Some SQL scripts required hotfixes mid-deployment to resolve issues with data transfers in our ETL process. Outside of our batch processing, analysts were now required to include new key fields in their scripts, or they would receive excessive amounts of data back. This resulted in abnormal SQL query times when running older versions of the queries. This also means whenever an analyst wants to add, update, or delete values in the database. The analyst had to ensure the new key fields were included in their scripts, or they could modify records across multiple tenants.

Selecting The Appropriate Data Analytics Tool or Technique

C. Describe the decision-making process of selecting the appropriate data analytics tool or technique from part B.

My current organization's decision-making process for selecting SQL lies in the fact that we support clients with a mid-to-large customer base where relational databases excel for data storage, organization, access, and processing. Utilizing SQL with a relational database within our organization allows analysts to better triage issues, perform analysis within the database, and run scripts/queries against the database to create data files for use with other tools outside of SQL. SQL allows this with easy-to-read language and quick execution time. Making it a great choice for most similar-sized organizations that want to provide easy access to data.

C1. Justify the organizational or technical need for the selected tool or technique.

From an organizational needs perspective, SQL is essentially the industry standard for database querying, significantly when your organization has already invested in deploying relational database schemas. There is no need to change our established and supported databases with SQL, whereas if we moved to a NoSQL tool, we would need to reevaluate our databases. SQL is also highly endorsed across multiple vendors, with no signs of SQL going anywhere soon. Many training materials exist to teach new employees how to utilize SQL for analytics work. This means a reduced training cost for the organization.

C2. Summarize the results of using the selected tool or technique in the life cycle phase you selected in B.

In the data acquisition phase, we use SQL queries with a combination of sub-queries and clauses to access the data our organization has stored in the database. We can then clean up some data with our queries to refine our output and better fit our analysis needs for the current project. This can involve removing duplicates, null values, or missing values. Once the query provides sufficient data, we can export the data in text, CSV, or TSV. Format and continue the DALC outside the database.

C3. Evaluate the 3 potential ethical problems of using the selected data analytics tool or technique identified in part B1 for this particular problem.

Catherine Cote (2001) summarizes it well:

> While the ethical use of data is an everyday effort, knowing that your data subjects' safety and rights are intact is worth the work. When handled ethically, data can enable you to make decisions and drive meaningful change at your organization and in the world.

Potential ethical programs with using SQL:

i) Data Privacy

   (1) SQL allows analysts to access databases containing sensitive personal data or confidential business information. The analyst may access sensitive personal data, such as health records, without authorization. This would be another privacy violation and a HIPPA violation, which could have legal ramifications for the organization. The mishandling of data when working with multiple clients could result in the analyst exposing one client's data to another. This would be a significant breach of privacy policies and regulations.

ii) Bias in decision-making

   (1) Analysts could use SQL queries to promote bias in their data models and visualizations by basing decisions on skewed outputs. While querying a dataset, the analyst may think it wise to group customers by zip code or phone number area code. While the dataset

may show specific trends, there is an unawareness of possible economic biases, which may be introduced to their data with cascading effects throughout the DALC.

iii) Lack of Consent

(1) Data analysts might access data without verifying customer consent or misuse the data outside the terms of use agreed upon by the customer. For example, some customers may opt out of allowing their data to be used for market purposes, so if an analyst were to use SQL to access the customer's data during the data acquisition phase and use it going forward in the DALC. The analyst has violated the customer's autonomy to a degree.

References

Cote, C. (2021, March 16). *5 principles of data ethics for business*. Business Insights Blog. https://online.hbs.edu/blog/post/data-ethics